Digitalizing Midwifery Care: AR and VR Solutions for Maternal Education

Project Proposal Report

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BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

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Declaration, copyright statement and the statement of the supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: .

Date: 8/29/2025

Abstract

The report consists of a comprehensive outline of how a Virtual Reality (VR) training environment and a postnatal support mobile application can modernize and improve the maternal healthcare sector by empowering midwives in delivering quality care. This VR training module focuses on newborn handling tasks such as carrying, cleaning, and breastfeeding guidance. The environment integrates interactive 3D models of infants, a virtual clinical space, and structured training sessions, creating an immersive, repeatable simulation for practical learning. Alongside the VR module, a mobile app is developed for midwives to manage postnatal care — enabling them to digitally record mother and baby vitals, generate reports, and receive automated reminders. The system ensures improved hands-on skills, better documentation, and accessible postnatal support.

The research gap involves the lack of integrated systems that combine immersive VR training with mobile-based postnatal care tools tailored specifically for midwives. Existing studies such as the *Virtual Midwife for Pregnant Women and Alert System* focus mainly on mobile alerts and communication, while the *Mobile-based VR Learning Environment for Postpartum Hemorrhage* highlights VR training for emergency situations rather than routine newborn care. Additionally, *A User-Centered Design of Natural Language Processing for Maternal Monitoring* supports mobile maternal tracking but lacks practical task integration and midwife-specific workflows. These studies point toward the need for a more complete, connected solution addressing both training and real-world support.

The novelty of this system lies in its dual-approach design: a VR-based training module for safe, interactive skill development, and a feature-rich mobile application focused on the postnatal stage of maternal care. The VR setup replicates a real clinical environment with interactive guidance and performance tracking, while the mobile app supports Sinhala voice-to-text, digital recordkeeping, and cloud-synced reports. References from IEEE research strengthen the foundation of this work, combining immersive learning and digital health tools into a unified system. Altogether, the project highlights the vital role of technology in advancing maternal care and midwifery education.

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List of Abbreviations

Abbreviation	Description
VR	Virtual Reality
AR	Augmented Reality
3D	3 Dimensional
EMS	Emergency Medical Service
API	Application Programming Interface
UI	User Interface
UX	User Experience
VS	Visual Studio
SDK	Software Development Kit
NHK	National Hospital Kandy

Introduction

Background & Literature survey

A fundamental competency in maternal and pediatric healthcare, particularly for midwives during and after childbirth, is newborn handling. These abilities include proper cleaning techniques, safe infant carrying techniques, nursing advice, and fundamental postnatal evaluations. During their clinical and academic training, midwifery students are expected to become proficient in these duties. However, many students lack sufficient practical experience prior to becoming certified experts because of restricted exposure to real-life situations and safety issues when handling babies.

Techniques like supporting the head and neck, washing without causing discomfort, and positioning for nursing are all part of proper infant handling. Every movement calls for attention and accuracy. Real-world conditions must be replicated in training settings so that midwives can learn by making mistakes and fixing them without endangering live babies. By providing trainees with a virtual 3D infant, a realistic clinical setting, and guided feedback, interactive virtual reality technology improves skill development through organized courses. Students can safely and regularly practice genuine events while interacting with lifelike models because to the immersive nature.

Figure 1: Safe Newborn Handling Positions in VR

After receiving their certification, midwives are in charge of managing and recording postpartum care. Manual data entry and conventional paper-based methods take a lot of time and are prone to mistakes. A secure mobile application designed especially for midwives is created to address this. With the use of this government-integrated program that allows for Sinhala voice-to-text input, midwives can easily collect mother and infant data. Smart vaccine and clinic appointment reminders, automated daily, monthly, and annual report generation, and a data analytics dashboard to monitor trends like average newborn weight by age are all included in the app. Previous studies have examined mobile health solutions, such as the Virtual Midwife Alert System [1] and the NLP-Based Maternal Monitoring System [2], but they have not integrated midwife-specific training and postnatal care elements.

In order to support the care of mothers and newborns, this study focuses on two different but complimentary improvements. As part of their pre-professional education, student midwives can safely practice critical infant handling methods in an immersive and interactive virtual reality (VR) training session. The second is a mobile application designed specifically for qualified midwives to handle the postpartum care period. It provides real-time analytics, smart notifications, automatic reporting, and secure voice-to-text data entry. When combined, these developments seek to enhance postnatal care quality and training efficacy in a contemporary healthcare setting.

Research Gap

Several research papers were reviewed in the domain of maternal and newborn care using VR and mobile-based applications. While all selected studies offered valuable insight into maternal health technologies, only a few directly addressed postnatal care tools for midwives or VR-based training for newborn handling. Most of the existing systems were either mobile-only with basic features or focused on emergency care rather than routine postnatal responsibilities.

Research A, "Virtual Midwife for Pregnant Women and Alert System", focuses on developing a mobile application for communication and emergency alerts during pregnancy. Although it supports general monitoring and notifications, it lacks dedicated features for structured postnatal care workflows, such as digital health tracking, data analytics, and official report generation used by midwives in the after-birth phase.

Research B, *"Mobile-based Virtual Reality Learning Environment for Postpartum Hemorrhage Management Training"*, presents a VR solution for training on maternal emergencies. It shows how VR can improve healthcare skills but does not cover routine

newborn care training or tasks that midwives must learn before certification, such as handling and breastfeeding guidance in the postnatal stage.

Research C, *"A User-Centered Design of NLP for Maternal Monitoring Chatbot System"*, explores the use of intelligent systems for maternal tracking through mobile chatbots. While it introduces useful voice-based interaction, it lacks Sinhala voice-to-text integration, field-level data analytics, and secure record-keeping designed for official midwife use. It is not aligned with the workflow standards required for postnatal care and reporting.

Table 1 Research Gap

Application Reference	Postnatal Workflow Coverage	VR Newborn Handling Training	Sinhala Voice-to-Text	Secure Recordkeeping	Data Analytics	Report Generation
Research A	x	x	x	x	x	x
Research B	x	✓	x	x	x	x
Research B	x	x	x	x	x	x
Our Proposed System	✓	✓	✓	✓	✓	✓

Research Problem

Learning how to handle a newborn safely can be challenging for midwifery students, especially during their training period where live practice opportunities are limited. Tasks like carrying, cleaning, and positioning a baby for breastfeeding require precision and confidence, but incorrect technique can pose risks. Without hands-on exposure, many students enter the workforce without fully developing these essential skills.

To address this, the proposed VR-based newborn care training environment gives midwifery students a safe and repeatable way to learn. The system offers interactive guidance on core postnatal tasks using realistic 3D newborn models and a virtual clinical setup. It simulates common scenarios midwives will face and allows them to practice with feedback, building real-world readiness through immersive repetition. Tasks such as holding the baby, supporting the neck, or positioning for breastfeeding are covered in structured modules.

On the other hand, practicing midwives face the challenge of keeping consistent digital records and managing after-birth care across many mothers. Manual data entry is time-consuming and increases the risk of missing critical follow-ups like vaccinations or clinic visits. Current systems also lack secure integration with local workflows and languages.

To solve this, a dedicated mobile application is being developed for midwives to manage after-birth care efficiently. The app supports Sinhala voice-to-text data entry, smart notifications for vaccine and clinic schedules, automated report generation, and data analytics (e.g., average child weight per age). It is designed with future government integration in mind, ensuring it can scale securely and support public health infrastructure. Together, both components target the real gaps in training and postnatal care management faced by midwives today.

Objectives

Main Objectives

The main objective of this project is to develop a realistic VR training environment for midwifery students to practice essential newborn handling techniques, and to build a secure mobile application for certified midwives to manage postnatal care efficiently.

VR Training Module : After wearing the VR headset and controllers, the midwifery student is placed in a virtual clinical room where various newborn care tasks can be practiced. The student can navigate to modules such as safe baby handling, cleaning, or breastfeeding assistance. Each task is guided through interactive prompts using a 3D animated newborn, allowing the learner to develop real-world skills in a safe, immersive space. Throughout the training, the student receives visual and audio feedback, ensuring mistakes can be corrected in real time and the learning is repeatable. The goal is to build confidence and accuracy before clinical placement.

Midwife Mobile App (After-Birth Phase) : The second component of this project is a mobile application designed specifically for government-authorized midwives. This app allows midwives to securely record mother and baby health information using Sinhala voice-to-text, minimizing the need for manual typing. The app sends automated reminders for vaccinations and clinic visits, and generates daily, monthly, and yearly reports of the midwife's activity. A data analytics dashboard helps track trends such as average infant weight across age groups. While not yet government-integrated, the system is built with the security and features required for future collaboration with public healthcare bodies.

Specific Objectives

VR Integration and Simulation Development

A fully immersive VR training environment will be developed using Unity Engine and OpenXR, enabling realistic simulation of newborn care procedures. This includes integrating interactive elements such as hand tracking and object interaction for tasks like carrying, cleaning, and positioning a newborn. The VR system serves as the core of the student training module; if this integration fails, the entire immersive learning experience would be compromised, making it the most critical technical objective.

3D Newborn Models and Clinical Environment Design

Realistic 3D models of newborns and a clinical postnatal care environment will be created to simulate hands-on procedures. Tools such as Blender or Maya will be used to model the virtual environment, which will include items like a hospital bed, infant cleaning station, and breastfeeding setup. Anatomical accuracy is essential, so references from maternal health research and neonatal care standards will guide the design. The objective

is to ensure realism so students can practice and build confidence in a safe yet life-like virtual setting.

VR User Interface (UI) and Task Flow Integration

An intuitive UI will be designed using Unity's VR toolkit to allow students to navigate through training modules, tutorials, and practice sessions. The system will include guided tasks and feedback-based challenges to reinforce learning. Voice prompts, interactive menus, and visual indicators will be embedded to help learners move through the training efficiently. Each task will track completion and errors, forming the foundation for assessment.

Mobile Application Development (After-Birth Phase)

A mobile app tailored for certified midwives will be developed using Flutter or Android Studio, focusing on postnatal care data management. This app will include secure login, Sinhala voice-to-text support, and cloud-based data storage. It will allow midwives to log mother and child information, send vaccination and clinic visit reminders, and auto-generate daily, monthly, and annual reports. Data analytics tools will also be embedded to track health trends like average infant weight by age group.

Data Security and Future Integration Readiness

Since the midwife app handles sensitive patient data, implementing high-level data security protocols is essential. This includes encrypted databases, role-based access controls, and compliance with health data protection standards. The system will be developed with a scalable architecture to support potential future integration with government health systems, pending official approval.

Methodology

The proposed system consists of two key components: a VR-based newborn care training module and a mobile application for certified midwives. Each component is developed using a combination of modeling tools, mobile frameworks, and immersive simulation engines to ensure a practical, user-friendly, and efficient experience for both student midwives and professional practitioners.

To begin with, the VR training module is developed using the Unity Engine alongside OpenXR for VR integration. This forms the foundation for immersive interaction. The clinical environment and newborn handling tasks are created using Blender, referencing real-world midwifery training practices. The 3D newborn models, clinical tools, and the simulation space are imported into Unity, where C# is used to program interactive controls, training tasks, and feedback systems. The result is a fully interactive learning lab

where students can perform safe handling, cleaning, and breastfeeding positioning in a risk-free setting.

The mobile application is developed separately using Flutter or Android Studio and is targeted for official use by government-approved midwives. The app includes a secure login, voice-to-text support in Sinhala, and allows the recording of mother and baby health information. Smart notifications are integrated to remind midwives of important tasks such as upcoming vaccinations or clinic checkups. The app generates daily, monthly, and annual activity reports and includes a data analytics dashboard that highlights trends, such as average infant weight across age ranges. Cloud database support is included for data backup and scalability.

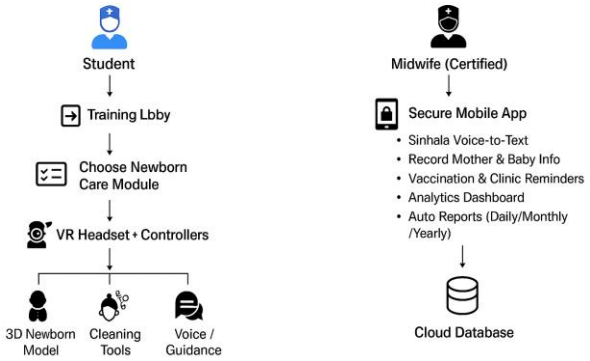

Figure 2: Individual System Diagram

Figure 2 shows the Individual System Diagram of the proposed solution, which includes both the VR Newborn Care Training Module and the Midwife Mobile Application. Initially, the student midwife is placed in a virtual training lobby or learning environment where they can select from a set of newborn care training modules. Once a module is selected, the student uses a VR headset running on Unity Engine with OpenXR integration, enabling them to interact with 3D newborn models, tools, and guided voice-based instructions in a realistic virtual environment.

On the other hand, the certified midwife accesses a secure mobile application designed for the after-birth care phase. Through this app, the midwife can record mother and baby information using Sinhala voice-to-text, set clinic and vaccination reminders, and generate automated daily, monthly, and yearly reports. All recorded data is synced to a cloud database, ensuring data integrity, analytics, and accessibility. Both systems are independent but serve a shared goal of improving maternal and newborn care.

The main component the VR environment consists of 5 different features or requirements, 3D models of equipment, and 3D heart model, various heart beat sounds, VR player control that gives the student intractability, and finally training, challenges and tasks all helping the student to train and learn using the environment. These all 5 features need a dataset of Cardiac Auscultation sounds, this includes various heart beat sounds and list of conditions that these sounds fit into.

Table 2 Technologies Used

Technology	**Description**
Unity Engine	Core development platform for VR training environment
OpenXR SDK	Used for VR hardware compatibility and immersive interaction
C# Language	Backend scripting and interaction logic in Unity
Blender / Maya	Tool for creating 3D newborn models and clinical environment
Flutter / Android Studio	Frameworks for building the mobile app for midwives
Firebase / Cloud DB	Cloud-based data storage and backup for mobile app
Google Speech API	For Sinhala voice-to-text conversion in mobile app
Visual Studio Code	Code editing and debugging environment

Functional Requirements

1. The VR training module should allow the user (midwifery student) to fully interact with the virtual environment, including selecting training modules and manipulating 3D newborn models.
2. Tasks such as safe carrying, cleaning, and breastfeeding guidance should function accurately, along with guided instructions and feedback mechanisms.
3. The mobile application should allow certified midwives to record data using Sinhala voice-to-text, send smart reminders, and generate automated reports.

Non-Functional Requirements

1. Medical Accuracy – All training tasks and postnatal care workflows must reflect up-to-date, verified medical procedures.

2. Model Accuracy – 3D newborn and tool models should be designed with realistic proportions and detail for effective training.
3. System Security – The mobile app should maintain a high level of data security to protect sensitive patient information.
4. User-Friendliness – Both the VR system and mobile app should offer an intuitive user interface for smooth navigation and ease of use.
5. Performance – The app and VR system should load quickly and operate smoothly without lag during training or data entry.

System Requirements

The VR module should include realistic audio-visual feedback (e.g., newborn responses, voice instructions), structured training modules, and performance tracking.
The mobile app must support real-time data recording, automated reminders, cloud syncing, and a dashboard for analytics and reporting.
Both components must operate independently but be designed for scalability and maintainability.

Hardware Requirements

- VR Headset (e.g., Oculus Quest, HTC Vive) with motion controllers
- High-performance PC capable of running Unity VR environments
- Smartphone or tablet for running the mobile midwife app
- Stable internet connection for cloud-based data syncing

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Software Requirements

- Unity Engine for developing the VR training module
- OpenXR SDK for VR device compatibility
- Blender / Maya / 3ds Max for modeling 3D newborns and clinical environments
- Flutter / Android Studio for mobile app development
- Firebase or equivalent for real-time database and cloud storage
- Google Speech API for Sinhala voice-to-text functionality

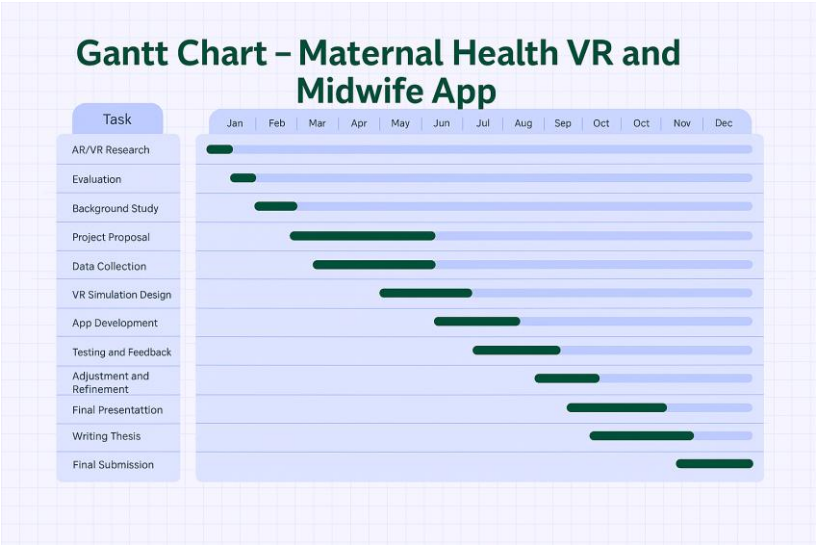

Figure 3 Gantt Chart

Description of Personal and Facilities

Supervisors

Our supervisor Mr. Didula Thanaweera Arachchi and Co-Supervisor Ms. Narmada Gamage, who are experts in the technical field will provide guidance in software development and technology integration.

External Supervisor

Guiding the healthcare aspect of the project is Senior Midwife Sandya Liyanage, who brings over 20 years of experience in maternal and postnatal care. Her extensive field expertise ensures the accuracy and practical relevance of both the VR training scenarios and the mobile application's features. Her input has been invaluable in shaping realistic training modules, improving usability for midwives, and validating clinical workflows used in the after-birth care phase.

Facilities

This research project is conducted in collaboration with the Humanities Department of SLIIT, with active involvement from nursing students. Through this partnership, access is provided to medical resources, nursing textbooks, and clinical guidelines. These facilities and references ensure that the developed VR training environment closely aligns with real-world midwifery practices, enhancing the clinical relevance and accuracy of the system

Sample of Medical Students and midwives

The project involves a sample group of medical and nursing students who will test the VR training module and provide detailed feedback. This offers them a valuable hands-on learning experience while contributing to system improvement. In addition, a group of certified midwives will test the mobile application, focusing on postnatal care tasks such as voice-to-text data entry, reminders, and report generation. Their feedback will be critical to ensuring the app meets real-world midwifery needs and follows practical workflows used in community healthcare settings.

High-Quality VR Equipment

For students to experience the environment as well as for testing purposes a high-quality VR headset and controllers are needed. These VR devices enables students to interact with the environment. Using a high-quality modern VR equipment provides access to the latest technology for medical purposes and simulation training.

Budget and Budget Justification

Table 3 Budget Table

Expenses	
Requirement	Cost (Rs.)
VR Meta Quest 3	186,000
Unity Engine License	57,000/month
Graphics Card (Nvidia RTX 3050)	97,500
Mobile Phone (Google Pixel 8 Pro)	200,000
Total Cost	540,500

Reference List

[1] R. Surantha, M. Zarlis, and D. Sitompul, "A User-Centered Design of Natural Language Processing for Maternal Monitoring Chatbot System," *2022 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS)*, 2022, pp. 1–6, doi: https://ieeexplore.ieee.org/document/10017517?utm=

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